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CSES Problem Set**Increasing Array II**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. You want to modify the array so that it is increasing, i.e., every element is at least as large as the previous element.

On each move, you can increase or decrease the value of any element by one. What is the minimum number of moves required?

Input

The first input line contains an integer n : the size of the array.

Then, the second line contains n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print the minimum number of moves.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input:

5
3 8 5 6 5

Output:

4

CSES Problem Set

Food Division

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n children around a round table. For each child, you know the amount of food they currently have and the amount of food they want. The total amount of food in the table is correct.

At each step, a child can give one unit of food to his or her neighbour. What is the minimum number of steps needed?

Input

The first input line contains an integer n : the number of children.

The next line has n integers $a_1, a_2, \dots, a_n$: the current amount of food for each child.

The last line has n integers $b_1, b_2, \dots, b_n$: the required amount of food for each child.

Output

Print one integer: the minimum number of steps.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $0 \leq a_i, b_i \leq 10^6$

Example

Input:

```
3
3 5 0
2 4 2
```

Output:

```
2
```

Explanation: Child 1 gives one unit of food to child 3, and child 2 gives one unit of food to child 3.

CSES Problem Set

Swap Round Sorting

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array containing a permutation of numbers $1, 2, \dots, n$, and your task is to sort the array using *swap rounds*. On each swap round, you can choose any number of distinct pairs of elements and swap each pair.

Your task is to find the minimum number of rounds and show how you can choose the pairs in each round.

Input

The first input line has an integer n : the size of the array.

The second line has n integers $x_1, x_2, \dots, x_n$: the initial permutation.

Output

First print an integer k : the minimum number of rounds.

Then, for each round, print the number of swaps and the indices of each swap. You can print any valid solution.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input:

```
5
5 2 1 3 4
```

Output:

```
2
2
1 3
4 5
1
3 5
```

Explanation: The initial array is $[5, 2, 1, 3, 4]$. After round 1, the array becomes $[1, 2, 5, 4, 3]$. After round 2, the array becomes $[1, 2, 3, 4, 5]$.

CSES Problem Set

School Excursion

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A group of n children are coming to Helsinki. There are two possible attractions: a child can visit either Korkeasaari (zoo) or Linnanmäki (amusement park).

There are m pairs of children who want to visit the same attraction. Your task is to find all possible alternatives for the number of children that will visit Korkeasaari. The children's wishes have to be taken into account.

Input

The first input line has two integers n and m : the number of children and their wishes. The children are numbered $1, 2, \dots, n$.

After this, there are m lines describing the children's wishes. Each line has two integers a and b : children a and b want to visit the same attraction.

Output

Print a bit string of length n where a one-bit at index i indicates that it is possible that exactly i children visit Korkeasaari (the bit string is to be considered one-indexed).

Constraints

- $1 \leq n \leq 10^5$
- $0 \leq m \leq 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 3
1 2
2 3
1 5
```

Output:

```
10011
```

Explanation: The number of children visiting Korkeasaari can be 1, 4 or 5.

CSES Problem Set

Coin Grid

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is an $n \times n$ grid whose each square is empty or has a coin. On each move, you can remove all coins in a row or column.

What is the minimum number of moves after which the grid is empty?

Input

The first input line has an integer n : the size of the grid. The rows and columns are numbered $1, 2, \dots, n$.

After this, there are n lines describing the grid. Each line has n characters: each character is either . (empty) or o (coin).

Output

First print an integer k : the minimum number of moves. After this, print k lines describing the moves.

On each line, first print 1 (row) or 2 (column), and then the number of a row or column. You can print any valid solution.

Constraints

- $1 \leq n \leq 100$

Example

Input:

```
3
..o
o.o
...
```

Output:

```
2
1 2
2 3
```

CSES Problem Set**Programmers and Artists**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A company wants to hire a programmers and b artists.

There are a total of n applicants, and each applicant can become either a programmer or an artist. You know each applicant's programming and artistic skills.

Your task is to select the new employees so that the sum of their skills is maximum.

Input

The first input line has three integers a , b and n : the required number of programmers and artists, and the total number of applicants.

After this, there are n lines that describe the applicants. Each line has two integers x and y : the applicant's programming and artistic skills.

Output

Print one integer: the maximum sum of skills.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $0 \leq a, b \leq n$
- $a + b \leq n$
- $1 \leq x, y \leq 10^9$

Example

Input:

```
2 1 4
3 7
9 8
1 5
4 2
```

Output:

```
20
```

Explanation: An optimal solution is to hire two programmers with skills 9 and 4 and one artist with skill 7. The sum of the skills is $9 + 4 + 7 = 20$.

CSES Problem Set

Coin Arrangement

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is a $2 \times n$ grid whose each cell contains some number of coins. The total number of coins is $2n$.

Your task is to arrange the coins so that each cell contains exactly one coin. On each move you can choose any coin and move it one step left, right, up or down.

What is the minimum number of moves if you act optimally?

Input

The first input line has an integer n : the width of the grid.

After this, there are two lines that describe the grid. Each line has n integers: the number of coins in each cell.

Output

Print one integer: the minimum number of moves.

Constraints

- $1 \leq n \leq 10^5$

Example

Input:

```
4
0 1 0 1
2 0 1 3
```

Output:

```
5
```

CSES Problem Set

Grid Puzzle I

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is an $n \times n$ grid, and your task is to choose from each row and column some number of squares. How can you do that?

Input

The first input line has an integer n : the size of the grid. The rows and columns are numbered $1, 2, \dots, n$.

The next line has n integers $a_1, a_2, \dots, a_n$: You must choose exactly a_i squares from the i th row.

The last line has n integers $b_1, b_2, \dots, b_n$: You must choose exactly b_j squares from the j th column.

Output

Print n lines describing which squares you choose (x means that you choose a square, . means that you don't choose it). You can print any valid solution.

If it is not possible to satisfy the conditions print only -1 .

Constraints

- $1 \leq n \leq 50$
- $0 \leq a_i \leq n$
- $0 \leq b_j \leq n$

Example

Input:

```
5
0 1 3 2 0
1 2 2 0 1
```

Output:

```
.....
..X..
.XX.X
XX...
.....
```

CSES Problem Set

Grid Puzzle II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is an $n \times n$ grid whose each square has some number of coins in it.

You know for each row and column how many squares you must choose from that row or column. You get all coins from every square you choose. What is the maximum number of coins you can collect and how could you choose the squares so that the given conditions are satisfied?

Input

The first input line has an integer n : the size of the grid. The rows and columns are numbered $1, 2, \dots, n$.

The next line has n integers $a_1, a_2, \dots, a_n$: You must choose exactly a_i squares from the i th row.

The next line has n integers $b_1, b_2, \dots, b_n$: You must choose exactly b_j squares from the j th column.

Finally, there are n lines describing the grid. You can assume that the sums of $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are equal.

Output

First print an integer k : the maximum number of coins you can collect. After this print n lines describing which squares you choose (X means that you choose a square, . means that you don't choose it).

If it is not possible to satisfy the conditions print only -1 .

Constraints

- $1 \leq n \leq 50$
- $0 \leq a_i \leq n$
- $0 \leq b_j \leq n$
- $0 \leq c_{ij} \leq 1000$

Example

Input:

```
5
0 1 3 2 0
1 2 2 0 1
2 5 1 5 1
0 2 5 1 2
3 8 9 3 5
1 4 3 7 3
0 3 6 2 8
```

Output:

```
32
.....
..X..
.XX.X
XX...
.....
```

CSES Problem Set**Bit Substrings**TASK [STATISTICS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

You are given a bit string of length n . Your task is to calculate for each k between $0 \dots n$ the number of non-empty substrings that contain exactly k ones.

For example, if the string is 101, there are:

- 1 substring that contains 0 ones: 0
- 4 substrings that contain 1 one: 01, 1, 1, 10
- 1 substring that contains 2 ones: 101
- 0 substrings that contain 3 ones

Input

The only input line contains a binary string of length n .

Output

Print $n + 1$ values as specified above.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input:

101

Output:

1 4 1 0

CSES Problem Set**Reversal Sorting**TASK [STATISTICS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

You have an array that contains a permutation of integers $1, 2, \dots, n$. Your task is to sort the array in increasing order by reversing subarrays. You can construct any solution that has at most n reversals.

Input

The first input line has an integer n : the size of the array. The array elements are numbered $1, 2, \dots, n$.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

First print an integer k : the number of reversals.

After that, print k lines that describe the reversals. Each line has two integers a and b : you reverse a subarray from position a to position b .

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input:

```
4
2 3 1 4
```

Output:

```
2
1 3
2 3
```

CSES Problem Set

Book Shop II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are in a book shop which sells n different books. You know the price, the number of pages and the number of copies of each book.

You have decided that the total price of your purchases will be at most x . What is the maximum number of pages you can buy? You can buy several copies of the same book.

Input

The first input line contains two integers n and x : the number of book and the maximum total price.

The next line contains n integers $h_1, h_2, \dots, h_n$: the price of each book.

The next line contains n integers $s_1, s_2, \dots, s_n$: the number of pages of each book.

The last line contains n integers $k_1, k_2, \dots, k_n$: the number of copies of each book.

Output

Print one integer: the maximum number of pages.

Constraints

- $1 \leq n \leq 100$
- $1 \leq x \leq 10^5$
- $1 \leq h_i, s_i, k_i \leq 1000$

Example

Input:

```
3 10
2 6 3
8 5 4
3 5 2
```

Output:

```
28
```

Explanation: You can buy three copies of book 1 and one copy of book 3. The price is $3 \cdot 2 + 3 = 9$ and the number of pages is $3 \cdot 8 + 4 = 28$.

CSES Problem Set

Mex Grid Queries

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Consider a two-dimensional grid whose rows and columns are 1-indexed. Each square contains the smallest nonnegative integer that does not appear to the left on the same row or above on the same column.

Your task is to calculate the value at square (y, x) .

Input

The only input line contains two integers y and x .

Output

Print one integer: the value at square (y, x) .

Constraints

- $1 \leq y, x \leq 10^9$

Example

Input:

3 5

Output:

6

CSES Problem Set

Stick Divisions

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You have a stick of length x and you want to divide it into n sticks, with given lengths, whose total length is x .

On each move you can take any stick and divide it into two sticks. The cost of such an operation is the length of the original stick.

What is the minimum cost needed to create the sticks?

Input

The first input line has two integers x and n : the length of the stick and the number of sticks in the division.

The second line has n integers $d_1, d_2, \dots, d_n$: the length of each stick in the division.

Output

Print one integer: the minimum cost of the division.

Constraints

- $1 \leq x \leq 10^9$
- $1 \leq n \leq 2 \cdot 10^5$
- $\sum d_i = x$

Example

Input:

```
8 3
2 3 3
```

Output:

```
13
```

Explanation: You first divide the stick of length 8 into sticks of length 3 and 5 (cost 8). After this, you divide the stick of length 5 into sticks of length 2 and 3 (cost 5). The total cost is $8 + 5 = 13$.

CSES Problem Set

Coding Company

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your company has n coders, and each of them has a skill level between 0 and 100. Your task is to divide the coders into teams that work together.

Based on your experience, you know that teams work well when the skill levels of the coders are about the same. For this reason, the penalty for creating a team is the skill level difference between the best and the worst coder.

In how many ways can you divide the coders into teams such that the sum of the penalties is at most x ?

Input

The first input line has two integers n and x : the number of coders and the maximum allowed penalty sum.

The next line has n integers $t_1, t_2, \dots, t_n$: the skill level of each coder.

Output

Print one integer: the number of valid divisions modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 100$
- $0 \leq x \leq 5000$
- $0 \leq t_i \leq 100$

Example

Input:

```
3 2
2 5 3
```

Output:

```
3
```